# 2017-Winter Paper 4 – Recent Advances

**Contents**


1. Q1. Robot and its role in urology (LAQ, 25)
2. Q2. Pathophysiology of allograft rejection; newer immunosuppressive drugs in renal transplant (LAQ, 25)
3. Q3a. Botulinum toxin in urology (SAQ, 10): cross-reference to 2017-Winter Paper 3
4. Q3b. Sacral neuromodulation (SAQ, 10): cross-reference to 2017-Winter Paper 3
5. Q3c. GnRH analogue (SAQ, 10)
6. Q3d. Miniperc (SAQ, 10)
7. Q3e. Andropause (SAQ, 10)
8. Q3f. Nephron saving surgery (SAQ, 10)


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## Q1. Discuss Robot and its role in urology.

**Type:** LAQ (25 marks)

### Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled.

### History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP in **2000**; first paediatric robotic procedure **2002**.

### da Vinci system

1. **Surgeon console**: 3D (three-dimensional) magnified view, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

| Generation | Added feature |
|---|---|
| S (2006) / Si | High definition, fourth arm, dual console for training |
| Xi (2014) | 8-mm camera in any port, finer arms, docks from either side, no stirrups |
| SP (2018) | All instruments through one 2.5-cm cannula |

- After patent expiry, other platforms appeared (e.g. **Hugo, Versius**, the Indian **SSI Mantra**).

### Advantages and disadvantages

| Advantages | Disadvantages |
|---|---|
| 3D vision, wristed instruments, easy suturing | High capital and instrument cost |

| Advantages | Disadvantages |
|---|---|
| Tremor filtration, seated ergonomics | Limited haptic feedback |
| Less blood loss and shorter stay than open; fewer conversions than laparoscopy | Docking time, bulk, surgeon away from patient |

## Applications (with evidence)

### 1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, less blood loss, shorter stay; equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable.
- **Retzius-sparing** and **hood** techniques preserve continence structures and give earlier continence.

### 2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy: shorter warm ischaemia and fewer conversions, so more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.

### 3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), less blood loss and transfusion.
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications.

### 4. Other procedures

| Procedure | Why the robot helps |
|---|---|
| Pyeloplasty, ureteric reimplantation | Easy intracorporeal suturing; also in children (small working space) |
| BMG (buccal mucosa graft) ureteroplasty | Long proximal or mid-ureteric strictures |
| RPLND (retroperitoneal lymph node dissection) | Template, nerve-sparing; short stay |
| Adrenalectomy, nephroureterectomy | Equivalent to laparoscopy |
| Simple prostatectomy | Large glands; less bleeding than open |
| Sacrocolpopexy, VVF (vesicovaginal fistula) repair | Deep pelvic suturing |
| **RAKT (robot-assisted kidney transplantation)** | Fewer wound complications in obese recipients |

## Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

## Physiology and positioning

- **$CO_2$ pneumoperitoneum** (12–15 mm Hg, lower in children) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery raises intracranial and intraocular pressure.

## Complications specific to robotics

| Problem | Details |
|---|---|
| Steep Trendelenburg | Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion |
| Nerve injury | Brachial plexus (avoid shoulder braces); femoral, peroneal |
| **Compartment syndrome, rhabdomyolysis** | Long lithotomy or flank position |
| Thermal bowel injury | Insulation failure, capacitive coupling; presents late |
| Robot malfunction | <1%; half found before the patient enters |
| Port-site hernia | Close larger port sites |

- **Prevention**: pad pressure points, secure arms before tilting, limit tilt and lithotomy time, protect the eyes.

## Training and credentialing

- Box trainer, **VR (virtual reality)** simulators, animal and cadaver labs, then proctored cases. **FRS (Fundamentals of Robotic Surgery)** curriculum. Outcomes improve with volume.

## Recent advances

- Single-port transvesical and extraperitoneal surgery; force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency).
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

**Diagram to draw:** console, vision cart and patient cart; RARP port placement.

**Key points to remember:**

- da Vinci 2000; Xi 2014; SP 2018; 7 degrees of freedom.
- Yaxley: RARP equal to open at 12 weeks; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.

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## Q2. Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant.

**Type:** LAQ (25 marks)

### Introduction

- Rejection is the recipient's immune response against donor (allo) antigens. Acute rejection has fallen from over 60% to **below 10%**; slow, subclinical **antibody injury** is now the main problem.

### Part A: Pathophysiology of rejection

#### Target antigens

- **HLA (human leucocyte antigen) / MHC (major histocompatibility complex)**: class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells.
- **ABO** antigens; minor and non-HLA antigens. **Sensitisation** (transfusion, pregnancy, previous transplant) creates preformed anti-HLA antibodies.

#### Allorecognition (how T cells see the graft)

| Pathway | Mechanism | Main role |
|---|---|---|
| **Direct** | Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells | Strong early acute TCMR (T-cell-mediated rejection) |
| **Indirect** | Recipient APCs (antigen-presenting cells) present processed **donor peptides on self-MHC** | Lifelong; helps B cells, so alloantibody and chronic rejection |
| **Semi-direct** | Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it | Links the two pathways |

- **Ischaemia–reperfusion injury** activates innate immunity and up-regulates MHC, so DGF (delayed graft function) raises rejection risk.

#### Three-signal model of T-cell activation

1. **Signal 1**: TCR (T-cell receptor) binds MHC–peptide → **calcineurin** → IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation)**: **CD28 binds CD80/86** (B7). Signal 1 without signal 2 causes anergy.
3. **Signal 3**: IL-2 binds **CD25** (IL-2 receptor) → **mTOR (mammalian target of rapamycin)** → G1 to S → proliferation (needs purine synthesis).

#### Effector mechanisms

- **Cellular**: CD8 cytotoxic T cells (perforin, granzyme) and CD4 Th1 cells with macrophages cause **tubulitis** and **endarteritis**.
- **Humoral**: **DSA (donor-specific antibody)** binds endothelium → complement, leaving **C4d** on peritubular capillaries → **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.

#### Types of rejection

| Type | Timing | Mechanism and features |
|---|---|---|
| **Hyperacute** | Minutes–hours | Preformed ABO/HLA antibody → complement, thrombosis; graft turns blue; **prevented by crossmatch** |

| Type | Timing | Mechanism and features |
|---|---|---|
| Accelerated acute | 2–5 days | Memory (anamnestic) response in sensitised recipients |
| **Acute TCMR** | Days–months (mostly first 6 months) | Interstitial infiltrate, tubulitis, arteritis |
| **Acute AMR** | Any time | DSA, C4d, microvascular inflammation |
| **Chronic active AMR** | Months–years | **Transplant glomerulopathy** (GBM (glomerular basement membrane) double contours), arterial intimal fibrosis; commonest cause of late graft loss |

- **Also know:** subclinical rejection (histology, stable function). **Risk factors**: HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant.

### Banff classification (biopsy is the gold standard)

| Category | Key criteria |
|---|---|
| Borderline | Foci of tubulitis with mild interstitial inflammation |
| TCMR IA / IB | Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis |
| TCMR IIA / IIB | Mild-moderate (IIA) or severe (IIB) intimal arteritis |
| TCMR III | Transmural arteritis or fibrinoid necrosis |
| Active AMR | All three: (1) tissue injury (microvascular inflammation, arteritis, TMA (thrombotic microangiopathy)); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA** |

### Diagnosis and monitoring

- Rising creatinine; **de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%) avoids unnecessary biopsy; gene-expression panels.
- **Exclude**: obstruction, leak, renal artery stenosis, CNI (calcineurin inhibitor) toxicity, BK virus nephropathy.

### Treatment

- **TCMR**: IV (intravenous) pulse steroids; **rATG (rabbit antithymocyte globulin)** for Banff II–III.
- **AMR** (not standardised): steroids, **IVIG (intravenous immunoglobulin)**, **plasmapheresis**, rituximab; eculizumab in selected patients.

## Part B: Newer immunosuppressive drugs

- Standard: induction plus **tacrolimus + mycophenolate** ± steroids. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

| Drug | Mechanism | Key points |
|---|---|---|
| **Belatacept** | **CTLA4-Ig**: binds CD80/86, blocks **signal 2** | IV monthly; **BENEFIT**: better GFR (glomerular filtration rate), more early rejection; **contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD (post-transplant lymphoproliferative disorder)) |
| **mTOR inhibitors** (sirolimus, everolimus) | Block mTOR (**signal 3**) | Not nephrotoxic; fewer CMV (cytomegalovirus) infections; **TRANSFORM**: everolimus + low-dose CNI non-inferior; poor wound healing, **lymphocele** |

| Drug | Mechanism | Key points |
|---|---|---|
| **Extended-release tacrolimus** | CNI (**signal 1**) | Once daily, better adherence |
| Basiliximab | Anti-CD25 (IL-2 receptor), non-depleting | Low-risk induction, days 0 and 4 |
| rATG; alemtuzumab | T-cell depletion; anti-CD52 | High-risk induction; infection, PTLD |
| **Rituximab; bortezomib** | Anti-CD20 (B cells); proteasome inhibitor (plasma cells) | AMR, desensitisation |
| **Eculizumab** | Anti-C5 | Selected AMR; aHUS (atypical haemolytic uraemic syndrome) |
| **Imlifidase** | Enzyme that **cleaves IgG** | EMA (European Medicines Agency) 2020: crossmatch-positive, highly sensitised |
| **Anti-IL-6** (tocilizumab, clazakizumab) | Blocks IL-6 | Chronic active AMR |

**Diagram to draw:** three-signal T-cell activation with drug sites (CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis).

**Key points to remember:**

- Direct pathway drives early TCMR; indirect pathway drives antibody and chronic rejection.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept: better GFR, avoid if EBV-negative; imlifidase for highly sensitised.

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## Q3a. Botulinum toxin in urology

**Type:** SAQ (10 marks)

**See 2017-Winter Paper 3, Q3b (full answer).**

**Key points to remember (recent advances angle):**

- Paediatric NDO (neurogenic detrusor overactivity): FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.
- Standard adult intradetrusor doses for comparison: onabotulinumtoxinA 100 U (idiopathic OAB (overactive bladder)), 200 U (NDO).

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## Q3b. Sacral Neuromodulation

**Type:** SAQ (10 marks)

**See 2017-Winter Paper 3, Q3a (full answer).**

**Key points to remember (recent advances angle):**

- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI (magnetic resonance imaging)-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

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# Q3c. GNRH Analogue

**Type:** SAQ (10 marks)

*Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.*

## Physiology

- **Pulsatile** GnRH stimulates pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone); LH drives Leydig-cell testosterone (about 95% of total). **Continuous** stimulation desensitises the receptor: the basis of agonist therapy.

## GnRH agonists

- **Drugs**: leuprolide, goserelin, triptorelin, histrelin. Position-6 substitution raises potency; depots last 1 month to 1 year.
- **Mechanism**: initial LH surge, then receptor **down-regulation** → low LH → castration in 2–4 weeks; survival equals orchidectomy.
- **Flare**: LH up to 10-fold for 10–20 days. Dangerous with spinal or weight-bearing metastases (cord compression, pain). **Prevent with an antiandrogen** (e.g. bicalutamide) started with or before the agonist, for **21–28 days**.

## GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC (subcutaneous) monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.
- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

## Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better. Check testosterone when PSA (prostate-specific antigen) rises.

## Indications in urology

- Metastatic, locally advanced (with radiotherapy) and recurrent prostate cancer; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty; cryptorchidism (GnRH or hCG (human chorionic gonadotrophin): low success, not first-line); endometriosis.

## Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction), **osteoporosis** (denosumab or zoledronic acid), diabetes, cardiovascular events, anaemia, sarcopenia.

**Diagram to draw:** hypothalamic–pituitary–testis axis showing agonist and antagonist sites.

**Key points to remember:**

- Agonist: surge, then down-regulation; cover flare with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE).
- Castrate <50 ng/dL; aim <20 ng/dL.

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# Q3d. Miniperc

**Type:** SAQ (10 marks)

## Definition

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.

| Designation | Tract size |
|---|---|
| Standard | 24–34 Fr |
| **Mini** | 15–22 Fr |
| Ultra-mini | 11–14 Fr |
| Micro-perc | <10 Fr (single-step "all-seeing needle") |

- The nomenclature is contested. The **working channel**, not sheath size, decides which instruments pass.

## Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Often **tubeless** or small nephrostomy (≤10 Fr: less pain).

## Advantages over standard PCNL

- Less haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; useful in children.

## Disadvantages

- **Longer operative time**, poorer visualisation, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a small sheath, especially when the scope nearly fills it, can raise pelvic pressure → pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

## Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding and stay, longer surgery; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears versus 24 Fr**.

## Indications

- Renal stones **1–2 cm**, **lower pole stones** (alternative to RIRS (retrograde intrarenal surgery)), failed SWL (shock wave lithotripsy), **paediatric stones**, calyceal diverticular stones; selected larger stones.
- **Contraindications**: as for PCNL (untreated infection, coagulopathy, pregnancy).

**Diagram to draw:** tract sizes compared (standard, mini, ultra-mini, micro).

**Key points to remember:**

- Mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998.
- Fewer transfusions and shorter stay, but longer operation.
- Watch intrarenal pressure; advantage lost against 24 Fr tracts.

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## Q3e. Andropause

**Type:** SAQ (10 marks)

*Read as: late-onset hypogonadism (testosterone deficiency of ageing men).*

### Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Andropause" is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

### Pathophysiology

- Mean testosterone in the seventh decade is about **one-third lower** than in young men.
- **Testis**: fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre**: blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so bioavailable (free + albumin-bound) testosterone falls more. Worse with obesity, diabetes, opioids.

### Clinical features

- **Sexual (most specific)**: low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood. **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

### Diagnosis

- **Morning (7–11 am) total testosterone, repeated.** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L); EAU (European Association of Urology): about 12 nmol/L. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone)**: high in primary, low or normal in secondary. Low LH → **prolactin**; pituitary MRI if testosterone <150 ng/dL or prolactin high.
- Baseline PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

### Treatment

- Lifestyle: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at about 3 months.
- **Contraindications**: prostate or breast cancer; **wish for fertility** (TRT suppresses spermatogenesis; use hCG (human chorionic gonadotrophin) or clomiphene); raised haematocrit; untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

### Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (hold if >54%), PSA.

### Evidence

- **TRAVERSE (2023)**: TRT non-inferior for major cardiac events; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

**Diagram to draw:** hypothalamic–pituitary–testis axis with ageing defects.

**Key points to remember:**

- Symptoms + low morning testosterone twice (AUA <300 ng/dL).
- SHBG rises with age; LH separates primary from secondary.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

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## Q3f. Nephron Saving Surgery (NSS)

**Type:** SAQ (10 marks)

### Definition

- Complete tumour removal with **maximal preservation of functioning parenchyma**, mainly PN (partial nephrectomy).
- **Rationale**: RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to cardiovascular events and death; PN gives equal cancer control with better function.

### Indications

| Category | Situations |
|---|---|
| Imperative | Solitary kidney, bilateral tumours, RN would mean dialysis |
| Relative | Contralateral kidney threatened (stone, stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma) |
| Elective | **cT1a with normal contralateral kidney: standard of care** |

- Increasingly used for cT1b–T2 when feasible. AUA (American Urological Association) 2021: RN only if high complexity, no CKD or proteinuria, and post-RN eGFR (estimated glomerular filtration rate) likely >45.

### Pre-operative assessment

- Multiphase CT (computed tomography) or MRI; **R.E.N.A.L. nephrometry score** (radius, exophytic, nearness to collecting system, anterior/posterior, location) predicts complexity.

### Technique (open, laparoscopic or robotic)

1. Expose kidney; **intraoperative ultrasound** maps the tumour.
2. Hilar control; clamp artery (± vein), selective, or off-clamp.
3. Excise with a thin rim, or **enucleate** along the pseudocapsule (familial, multifocal tumours).
4. Close collecting system and vessels; renorrhaphy; unclamp; drain.

- **Robotic PN**: shorter warm ischaemia and fewer conversions than laparoscopy.

### Ischaemia

- Aim warm ischaemia **<25–30 minutes**; **cold ischaemia (ice slush)** if longer is expected, especially in a solitary kidney.
- Function depends mainly on **parenchyma preserved**. Randomised trials: no functional benefit of off-clamp, hypothermia or mannitol with a normal contralateral kidney.
- **Trifecta**: negative margin + no major complication + warm ischaemia ≤25 minutes.

### Complications

- **Urine leak** (stent, drain), **bleeding/pseudoaneurysm** (angioembolisation), AV (arteriovenous) fistula, renal insufficiency, hyperfiltration proteinuria (ACE (angiotensin-converting enzyme) inhibitors).

### Outcomes

- Local recurrence 1–2% (elective), 3–5% (imperative). Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011)**: RN vs PN for ≤5 cm: PN better function, similar oncology; overall survival slightly

better with RN, but trial flawed.

- **Also know:** thermal ablation for <3 cm; surveillance in frail patients.

**Diagram to draw:** tumour excised with rim, clamp on renal artery, two-layer renorrhaphy.

> **Key points to remember:**
>
> - Imperative, relative, elective; elective PN is standard for cT1a.
> - Warm ischaemia ≤25 min; preserved parenchyma matters most.
> - EORTC 30904: PN better function, similar cancer control.

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# Rapid Revision Sheet – 2017-Winter Paper 4

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Discuss Robot and its role in urology. (LAQ)**

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**Q2. Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant. (LAQ)**

- Direct pathway drives early TCMR; indirect pathway drives antibody and chronic rejection.
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